

A Taxonomy of Hallucination Types in Large Language Models for Research Applications

Abstract

Large Language Models (LLMs) have rapidly become important research assistants for literature discovery, scientific question answering, summarization, hypothesis generation, coding, data interpretation, and scholarly writing. Their usefulness, however, is constrained by a fundamental reliability problem: they can generate fluent, plausible, and well-structured statements that are unsupported, unverifiable, or factually incorrect. This phenomenon is commonly referred to as *hallucination*. Existing literature has established that hallucination is not a single failure mode. It can arise from deficiencies in parametric knowledge, conflicts with provided evidence, failures to follow instructions, temporal knowledge gaps, retrieval errors, reasoning failures, citation problems, and inappropriate expressions of uncertainty. Earlier taxonomies developed for task-specific natural language generation distinguish primarily between intrinsic and extrinsic hallucination, whereas recent LLM research emphasizes broader distinctions involving factuality and faithfulness.

This paper develops a research-oriented taxonomy of hallucinations specifically for scholarly and scientific applications. The proposed taxonomy organizes hallucinations along four complementary dimensions: **factual grounding, evidence faithfulness, research provenance, and reasoning/epistemic behavior**. Ten practically relevant categories are identified: fabricated facts, entity and relation errors, unsupported inference, contradiction of supplied evidence, context omission/distortion, fabricated citations and bibliographic metadata, source-attribution errors, temporal hallucination, quantitative/mathematical hallucination, and reasoning-induced hallucination. The paper further examines current detection and mitigation approaches, including retrieval-augmented generation, citation-aware generation, atomic factuality evaluation, self-consistency methods, semantic-entropy estimation, verification chains, and post-generation research-and-revision systems. Particular attention is given to why these techniques remain insufficient for research settings, where correctness must be accompanied by traceability, reproducibility, source quality, uncertainty calibration, and temporal validity. Finally, the paper identifies research gaps concerning multidimensional hallucination benchmarks, citation-level verification, domain-specific factuality, uncertainty-aware scholarly agents, multimodal scientific hallucination, and standardized evaluation protocols. The proposed taxonomy is intended as a conceptual framework for developing more reliable LLM-based research assistants and for designing future datasets, metrics, and mitigation systems.

Keywords: Large Language Models, LLM hallucination, hallucination taxonomy, factuality, faithfulness, scientific AI, research assistants, citation verification, retrieval-augmented generation, trustworthy AI, factuality evaluation

1. Introduction

Large Language Models (LLMs) have transformed natural language processing from task-specific prediction systems into general-purpose computational assistants capable of question answering, summarization, information retrieval, code generation, reasoning, translation, and document analysis. Their increasing adoption in research environments has created opportunities to automate or accelerate substantial

portions of the scholarly workflow, including literature exploration, manuscript drafting, technical explanation, programming assistance, experimental analysis, and synthesis of findings.

However, the fluency of an LLM response does not guarantee its factual correctness. A model may generate a statement that is syntactically coherent and semantically plausible while lacking adequate evidence or directly contradicting established knowledge. This phenomenon is generally described as hallucination. The problem was investigated before the emergence of modern LLMs in neural natural language generation, where hallucination was commonly associated with generated content that was either inconsistent with a source document or not supported by it (Ji et al., 2023). With instruction-tuned LLMs, the phenomenon has become broader because models are no longer restricted to source-grounded generation and routinely answer open-domain questions from their internal parametric representations.

The distinction between factuality and hallucination is important. Factuality concerns whether a generated claim is correct with respect to an appropriate external standard of truth, whereas faithfulness concerns whether a response appropriately reflects the information, instructions, or evidence provided to the model (Wang et al., 2024; Huang et al., 2025). An answer can therefore be factually correct but unfaithful to a supplied source, or factually incorrect despite appearing internally consistent.

Research applications make this distinction particularly consequential. A hallucinated sentence in an informal conversation may be inconvenient; a fabricated DOI, nonexistent paper, incorrect experimental result, or invented statistical value in a scientific manuscript can contaminate the scholarly record. Similarly, a citation that appears legitimate but does not actually support the accompanying claim can create a false chain of evidence. The problem is particularly serious because users may overestimate the reliability of highly fluent systems.

Empirical benchmarks demonstrate the persistence of this problem. TruthfulQA showed that contemporary language models could reproduce common misconceptions and false beliefs, with the best evaluated model answering truthfully only 58% of benchmark questions in the original study (Lin et al., 2022). HaluEval subsequently demonstrated that LLMs can generate and also fail to recognize hallucinated content, while external knowledge and additional reasoning can improve hallucination recognition (Li et al., 2023). More fine-grained work such as FActScore demonstrated that long-form generations frequently contain mixtures of supported and unsupported atomic facts, making binary document-level factuality judgments inadequate (Min et al., 2023).

For research applications, therefore, a useful taxonomy must go beyond the binary question of whether an answer is "hallucinated." It should identify **what is wrong, what evidence is violated or absent, why the failure occurred, and what type of verification is required.**

This paper has four objectives:

1. to review established conceptualizations of hallucination in NLG and LLMs;
2. to develop a taxonomy specifically oriented toward research and scholarly applications;
3. to examine contemporary hallucination detection and mitigation approaches; and
4. to identify open research problems for trustworthy LLM-based research systems.

2. Background and Related Work

2.1 Hallucination in Natural Language Generation

Hallucination has a substantial history in neural text generation. Ji et al. (2023) provide a comprehensive survey of hallucination in NLG and distinguish between **intrinsic** and **extrinsic** hallucinations. Intrinsic hallucination occurs when generated information contradicts the source input, whereas extrinsic hallucination refers to generated information that cannot be verified from the source. This distinction was particularly useful for summarization and data-to-text generation.

The emergence of general-purpose LLMs has expanded the problem. LLMs are frequently used without an explicit source document and can produce responses based on their parametric knowledge. Consequently, a hallucination taxonomy designed exclusively around source-grounded NLG does not adequately describe modern research-assistant behavior.

Recent surveys have consequently proposed more LLM-specific taxonomies. Zhang et al. (2023) characterize hallucinations in terms of divergence from user input, contradictions with previously generated context, and conflicts with established world knowledge. A later comprehensive survey by Huang et al. (2025) organizes LLM hallucinations primarily into **factuality hallucination** and **faithfulness hallucination**, emphasizing that LLM hallucination has become broader than traditional source-conditioned NLG.

Wang et al. (2024) similarly emphasize that factuality, hallucination, relevance, and trustworthiness are related but distinct evaluation dimensions. Their survey highlights the difficulty of automatically evaluating factuality in open-ended LLM generation and reviews retrieval, self-reflection, and other strategies for improving factual reliability.

2.2 Factuality Versus Faithfulness

A useful conceptual distinction is:

$$\text{Factuality} = \text{Agreement with verifiable external reality}$$

whereas

$$\text{Faithfulness} = \text{Agreement with the information and constraints supplied in context.}$$

Suppose a researcher supplies a paper stating that an algorithm achieves 92% accuracy, while the LLM responds that it achieves 95%. This is a **faithfulness failure** because the response contradicts the provided source. If the source itself contains an incorrect result, however, the response may be faithful to the source while being factually wrong in the broader world.

Research systems therefore require both dimensions simultaneously.

2.3 Hallucination and Truthfulness

Truthfulness is closely related but not identical to factuality. TruthfulQA was designed specifically to test whether models reproduce human misconceptions rather than merely optimize likelihood over common

textual patterns. Lin et al. (2022) found that models could produce persuasive false answers, demonstrating that language-model fluency can coexist with poor truthfulness.

This observation is especially important for scholarly applications because scientific writing has strong conventions of authority and precision. A fabricated claim accompanied by formal academic language can appear more credible than an obviously uncertain answer.

2.4 Hallucination as an Epistemic Problem

An important dimension of hallucination is not simply *whether the model is wrong*, but *whether the model appropriately represents its uncertainty*. Kadavath et al. (2022) showed that language models possess some capacity to estimate whether their answers are likely to be correct, although calibration and generalization remain imperfect.

Thus, a reliable research assistant should ideally distinguish among:

- **known and supported information,**
- **plausible but uncertain information,**
- **information requiring external verification, and**
- **information for which the model has insufficient evidence.**

The failure to make these distinctions can itself constitute a research-oriented hallucination.

3. Proposed Taxonomy of Hallucination Types for Research Applications

Existing taxonomies are valuable but are often organized according to either generation architecture or general LLM behavior. For research applications, a more operational taxonomy is required.

This paper proposes four high-level dimensions:

1. **Factual Grounding**
2. **Evidence and Context Faithfulness**
3. **Research Provenance**
4. **Reasoning and Epistemic Reliability**

These dimensions produce ten principal hallucination types.

Category	Hallucination type	Core failure	Typical research example
Factual grounding	Fabricated fact	Invented or false claim	Incorrect scientific statement

Category	Hallucination type	Core failure	Typical research example
Factual grounding	Entity/relation error	Wrong entity, attribute, or relationship	Wrong author/year/algorithm association
Factual grounding	Temporal hallucination	Outdated or future-inconsistent knowledge	Incorrect current software/version information
Evidence faithfulness	Source contradiction	Output conflicts with supplied evidence	Wrong interpretation of a paper
Evidence faithfulness	Context omission/distortion	Relevant evidence is ignored or selectively represented	Missing study limitation
Evidence faithfulness	Unsupported inference	Conclusion exceeds evidence	Causal claim from correlational result
Research provenance	Citation fabrication	Nonexistent reference or metadata	Invented DOI/paper
Research provenance	Citation-support hallucination	Real citation does not support claim	Citation laundering
Reasoning	Quantitative/mathematical hallucination	Incorrect calculation or numerical claim	Wrong statistical result
Reasoning/epistemic	Reasoning-induced or confidence hallucination	Plausible reasoning leading to false conclusion or unjustified certainty	Confidently incorrect theorem interpretation

This taxonomy should be regarded as **multidimensional rather than mutually exclusive**. A single generated sentence may exhibit several categories simultaneously.

3.1 Type I: Fabricated Factual Claims

The most recognizable hallucination occurs when an LLM generates a proposition that is not supported by reality.

For example:

"The Transformer architecture was introduced in 2018 by Google."

The statement is plausible but contains factual errors regarding both year and attribution. Such errors are particularly dangerous in literature reviews because they can propagate incorrect historical narratives.

Fabricated facts can involve:

- dates,
- scientific discoveries,
- algorithms,
- experimental findings,
- institutions,
- researchers,
- datasets,
- software packages,
- performance results, and
- technical specifications.

FactScore provides an important methodological response to this problem by decomposing long-form text into atomic claims and measuring the fraction supported by reliable evidence (Min et al., 2023). The study demonstrates why document-level judgments can conceal substantial variation in factual reliability.

3.2 Type II: Entity and Relation Hallucinations

Some hallucinations do not invent an entire fact but instead modify an entity or relationship.

Examples include:

- attributing a paper to the wrong author;
- assigning an algorithm to the wrong researcher;
- confusing two datasets;
- associating a DOI with the wrong article;
- claiming that a method was introduced by an institution that did not develop it.

Formally, if a knowledge graph represents a fact as

$$(e_1, r, e_2),$$

then an entity hallucination may replace e_1 or e_2 , while a relation hallucination changes r .

For research systems, this category is particularly important because bibliographic knowledge is highly relational. A sentence may contain individually plausible entities but an incorrect relationship between them.

3.3 Type III: Temporal Hallucinations

LLMs are trained on datasets collected at particular points in time. Consequently, their parametric knowledge can become outdated.

Temporal hallucinations occur when the model:

1. reports obsolete information as current;
2. fails to incorporate a known change;
3. incorrectly predicts a historical state;
4. attributes a current capability to an earlier system; or
5. produces anachronistic claims.

FreshLLMs and the FreshQA benchmark specifically demonstrate the difficulty of answering questions involving rapidly changing knowledge. Their evaluation found that LLMs struggle with fast-changing facts and false-premise questions and showed that search augmentation can substantially improve performance.

For research applications, temporal hallucination affects:

- software versions,
- research funding schemes,
- institutional leadership,
- standards,
- datasets,
- scientific discoveries,
- clinical recommendations,
- regulations, and
- rapidly evolving research fields.

Thus, "knowledge correctness" should include a temporal dimension:

$$C = C_{\text{factual}} \wedge C_{\text{temporal}}.$$

3.4 Type IV: Source-Contradiction Hallucination

Source contradiction occurs when an LLM explicitly or implicitly contradicts information supplied in the prompt or retrieved context.

Suppose a research paper states:

Accuracy = 87.4%

but the model summarizes:

Accuracy = 91.2%.

The generated statement may appear scientifically plausible, but it is inconsistent with the supplied evidence.

This category corresponds closely to intrinsic hallucination in traditional NLG and to faithfulness hallucination in contemporary LLM research (Ji et al., 2023; Huang et al., 2025).

Source contradiction is especially relevant to:

- paper summarization,
- systematic reviews,
- experimental result extraction,
- thesis assistance,
- legal research,
- scientific QA, and
- document-grounded agents.

3.5 Type V: Context Omission and Distortion

A model may avoid making a directly false statement while still producing a misleading representation of the source.

For example, a scientific paper may report:

"The method improves classification accuracy under the tested dataset and experimental conditions but has substantial performance degradation under domain shift."

A generated summary may state only:

"The method significantly improves classification accuracy."

The second statement may not be literally false, but omission of the limitation can materially distort the scientific conclusion.

This category demonstrates why factuality alone is insufficient. Scholarly reliability requires preservation of **scope, qualifiers, assumptions, limitations, and uncertainty**.

3.6 Type VI: Unsupported Inference

Unsupported inference occurs when a model derives a conclusion that is not justified by the available evidence.

A common scientific example is:

$$X \text{ correlates with } Y \Rightarrow X \text{ causes } Y.$$

The first proposition may be supported while the second is not.

Other examples include:

- inferring generalization from one dataset;
- claiming statistical significance without a test;
- claiming superiority from a single metric;
- inferring mechanism from association;
- interpreting observational evidence as causal;
- claiming reproducibility without independent replication.

This category is particularly relevant because modern LLMs can produce extensive chains of reasoning. Longer reasoning does not necessarily imply stronger evidence.

3.7 Type VII: Citation Fabrication

Citation hallucination is one of the most serious forms of research hallucination.

It occurs when the model invents:

- nonexistent papers,
- fake authors,
- fabricated titles,
- incorrect journal names,
- nonexistent DOIs,
- false publication years,
- incorrect page numbers, or
- combinations of real entities that do not correspond to a real publication.

Citation fabrication differs from ordinary factual hallucination because the fabricated information is presented as a **verification mechanism**.

A researcher may therefore mistakenly treat the citation as evidence that validates the claim.

The problem motivates citation-aware generation research. Gao et al. (2023), for example, introduced ALCE, a benchmark for evaluating LLM-generated citations along dimensions including fluency, correctness, and citation quality. Their experiments showed substantial remaining deficiencies in citation completeness.

3.8 Type VIII: Citation-Support and Attribution Hallucination

A citation can be real and still be hallucinated in a functional sense if it does not support the claim attached to it.

Consider:

"Quantum machine learning provides exponential speedup in practical classification tasks [Smith et al., 2023]."

Suppose the cited paper is genuine but only presents a theoretical framework and does not establish an exponential practical speedup. The reference exists, but the **citation-to-claim relationship is false**.

This can be represented as:

$$\text{Claim} \xrightarrow{\text{citation}} \text{Source}$$

where hallucination occurs if

$$\text{Support}(\text{Claim}, \text{Source}) \approx 0.$$

Therefore, citation verification must operate at the claim-source level rather than merely checking whether the cited paper exists.

ALCE explicitly addresses this issue by evaluating citation correctness and citation quality rather than simply citation presence (Gao et al., 2023).

RARR similarly approaches attribution as a post-generation research and revision problem, automatically finding supporting evidence and revising unsupported content (Gao et al., 2023).

3.9 Type IX: Quantitative and Mathematical Hallucination

LLMs can produce incorrect numerical results even when the surrounding explanation appears sophisticated.

Examples include:

- incorrect percentages;
- fabricated benchmark scores;
- incorrect statistical tests;
- wrong confidence intervals;
- incorrect equations;
- arithmetic errors;
- invalid dimensional analysis;
- fabricated complexity results; and
- incorrect parameter values.

Research applications are particularly vulnerable because numerical claims frequently carry more evidentiary weight than qualitative statements.

A useful distinction is:

$$\text{Numerical hallucination} = \text{incorrect numerical value}$$

versus

quantitative reasoning hallucination = invalid procedure producing a plausible value.

The second is often harder to detect because the answer may appear mathematically coherent.

3.10 Type X: Reasoning-Induced and Confidence Hallucination

The final category concerns hallucinations generated through reasoning or expressed with inappropriate confidence.

A model may start with incomplete information, introduce an unsupported assumption, and then produce a logically coherent chain leading to a false conclusion.

The problem can be represented as:

Incomplete premise → Unsupported assumption → Coherent reasoning → False conclusion.

This is distinct from a simple factual error because the final answer may contain several individually plausible intermediate steps.

Confidence hallucination occurs when the model communicates an unsupported proposition with excessive certainty.

For scientific systems, epistemic calibration is therefore essential. A research assistant should be capable of saying:

- "The source establishes X."
 - "This interpretation is plausible but not directly demonstrated."
 - "The available evidence is insufficient."
 - "I could not verify this citation."
 - "This claim requires independent verification."
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4. Hallucination in Research Workflows

Hallucination types become easier to understand when mapped onto the research lifecycle.

4.1 Literature Discovery

During literature search, hallucinations may involve nonexistent papers, incorrect titles, incorrect authors, or inappropriate recommendations.

4.2 Literature Review

During synthesis, models may merge findings from different papers, attribute results incorrectly, or omit important limitations.

4.3 Methodology Development

A model may recommend a method based on an incorrect assumption about a dataset, algorithm, or experimental setting.

4.4 Experimental Analysis

Hallucinations may appear as fabricated metrics, incorrect statistical interpretation, or unsupported claims of improvement.

4.5 Scholarly Writing

Citation fabrication, citation-support errors, and overclaiming become major risks.

4.6 Peer-Review Assistance

An LLM may invent weaknesses, attribute statements to a paper that it does not contain, or incorrectly assess novelty.

Consequently, hallucination risk should not be considered a single model-level property. It is better represented as:

$$H = f(M, D, T, C, R, U),$$

where:

- M = model characteristics,
 - D = domain,
 - T = task,
 - C = contextual evidence,
 - R = retrieval quality, and
 - U = user interaction/prompt conditions.
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5. Current Approaches for Hallucination Detection and Mitigation

5.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines parametric language models with external information retrieval. Its central motivation is to provide the model with evidence that is more current and task-specific than its internal parameters.

RAG can be viewed as:

$$q \rightarrow \text{Retriever}(q) \rightarrow D \rightarrow \text{Generator}(q, D) \rightarrow y.$$

The approach can reduce errors arising from missing or outdated parametric knowledge. However, RAG does not eliminate hallucination.

RAGTruth demonstrates that LLMs can still generate unsupported or contradictory claims despite retrieved evidence. The dataset contains nearly 18,000 naturally generated RAG responses with manual annotations at case and word levels and shows that retrieval-augmented systems continue to exhibit hallucination.

Thus:

$$\text{RAG} / \text{guaranteed factuality.}$$

A more realistic model is:

$$P(H) = f(Q, R, G),$$

where Q represents query quality, R retrieval quality, and G generation behavior.

5.2 Citation-Aware Generation

Citation-aware systems attempt to associate generated claims with supporting evidence.

ALCE introduced a benchmark in which systems retrieve supporting documents and generate answers containing citations, with evaluation of fluency, correctness, and citation quality (Gao et al., 2023).

The research implication is significant: citation generation should not be evaluated only on whether references are present. A robust system must evaluate:

1. citation existence;
2. citation validity;
3. citation relevance;
4. claim-source entailment;

- 5. citation completeness; and
 - 6. source quality.
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5.3 Atomic Factuality Evaluation

FactScore decomposes a long-form response into atomic facts and evaluates whether those facts are supported by a reliable knowledge source (Min et al., 2023).

Conceptually:

$$FactScore = \frac{\#supported\ atomic\ facts}{\#atomic\ facts}.$$

This approach is particularly valuable for research because scholarly responses frequently contain dozens of factual claims.

However, factual support alone does not measure:

- source quality,
 - scientific importance,
 - reasoning validity,
 - citation completeness,
 - uncertainty,
 - novelty,
 - methodological appropriateness, or
 - causal validity.
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5.4 Self-Consistency and Sampling-Based Detection

SelfCheckGPT uses stochastic sampling from a black-box LLM and measures whether independent generations remain semantically consistent. The intuition is that known information tends to generate more consistent responses, whereas hallucinated information may vary across samples (Manakul et al., 2023).

The method is attractive because it does not require access to model probabilities or an external database.

However, self-consistency is not equivalent to truth. A model can consistently repeat the same incorrect fact. Therefore:

$$\text{Consistency} \not\Rightarrow \text{Truth}.$$

This is a fundamental limitation for scholarly applications.

5.5 Semantic Entropy and Uncertainty Estimation

Farquhar et al. (2024) introduced semantic entropy for detecting hallucinations by measuring uncertainty over semantic equivalence classes rather than raw token probabilities. Their Nature study demonstrated that semantic uncertainty can be useful for detecting unreliable outputs, including on questions where the correct answer is not known to the evaluator in advance.

This approach is promising because a research assistant could potentially abstain when uncertainty is high:

$$\text{if } U(y) > \tau, \quad \text{request verification.}$$

However, confidence-based methods face an important problem: models can be highly confident in incorrect answers. Therefore, calibration remains an unresolved issue.

5.6 Chain-of-Verification

Dhuliawala et al. (2023) proposed Chain-of-Verification (CoVe), in which the model drafts an answer, creates verification questions, independently answers those questions, and then produces a revised answer. Experiments showed reductions in hallucination across multiple tasks.

For research applications, the method suggests an important design principle:

$$\text{Generate} \rightarrow \text{Question} \rightarrow \text{Verify} \rightarrow \text{Revise.}$$

Nevertheless, if the model itself generates the verification evidence, the process may remain vulnerable to correlated errors. Independent external verification is therefore preferable for high-stakes scholarly claims.

5.7 Research-and-Revision Systems

RARR extends this concept by using external research to find attribution for model-generated statements and revise unsupported content while preserving the original response as much as possible (Gao et al., 2023).

Such systems are particularly relevant to scholarly workflows because they shift hallucination mitigation from:

"generate correctly the first time"

to:

"generate $\rightarrow$ investigate $\rightarrow$ verify $\rightarrow$ revise."

The latter is closer to the actual scientific method.

6. Comparative Analysis of Current Approaches

Approach	Main target	External evidence	Strength	Major limitation
Self-consistency	Inconsistency	No	Black-box compatible	Consistent errors remain possible
FactScore	Atomic factuality	Yes	Fine-grained evaluation	Depends on knowledge source
ALCE	Citation correctness	Yes	Research-oriented	Retrieval quality affects results
RAG	Knowledge grounding	Yes	Reduces knowledge gaps	Retrieved evidence can be wrong/irrelevant
RARR	Attribution + revision	Yes	Corrects unsupported claims	Computationally expensive
CoVe	Self-verification	Optional	Reduces several hallucinations	Verification can share model errors
Semantic entropy	Epistemic uncertainty	No	Can detect uncertainty without reference answer	Calibration remains difficult
RAGTruth	RAG hallucination evaluation	Yes	Word-level annotations	Focused primarily on RAG
FreshQA/ FreshPrompt	Temporal factuality	Search	Addresses dynamic knowledge	Search quality and freshness remain factors

The evidence indicates that no single approach solves hallucination comprehensively. This supports a **defense-in-depth architecture** rather than a single detector.

7. Research Challenges and Limitations

7.1 No Universal Definition of Hallucination

The literature does not use a single operational definition. Terms such as *hallucination*, *factuality error*, *faithfulness error*, *misinformation*, *unsupported claim*, and *confabulation* overlap but are not identical.

A universal evaluation framework must therefore explicitly specify the reference standard.

7.2 Truth Is Often Difficult to Operationalize

Scientific claims can depend on:

- publication date,
- experimental conditions,
- population,
- dataset,
- statistical assumptions,
- measurement procedures, and
- competing theories.

Consequently, "truth" may require contextual qualification.

7.3 Citation Verification Is More Difficult Than Citation Detection

Checking whether a DOI exists is comparatively easy. Determining whether the cited article actually supports a specific claim is considerably harder.

A robust scholarly verifier should evaluate:

$$P(\text{Source supports claim} \mid C, S),$$

where C is the claim and S is the source.

This requires semantic entailment, source interpretation, and often multi-document reasoning.

7.4 Long-Context Models Do Not Automatically Solve Evidence Grounding

Longer context windows allow models to ingest more scientific papers, but greater context length does not guarantee effective information use. Liu et al. (2024) showed that model performance can substantially deteriorate when relevant information is positioned in the middle of long contexts, producing a characteristic U-shaped performance pattern.

For research agents, simply placing an entire literature corpus into the context window is therefore not a reliable substitute for effective retrieval and evidence selection.

7.5 Domain-Specific Hallucination

A model can perform well on general factuality benchmarks while failing in specialized domains such as:

- quantum computing,
- remote sensing,
- biomedical research,
- law,
- telecommunications,
- climate science, or
- advanced mathematics.

Domain-specific terminology increases the probability of entity and relation errors, while specialized scientific claims may have relatively few examples in the model's training distribution.

7.6 Multimodal Hallucination

Scientific research increasingly depends on:

- satellite imagery,
- medical images,
- plots,
- diagrams,
- tables,
- equations,
- experimental apparatus, and
- geospatial data.

Hallucination can therefore occur across modalities. An LLM may correctly interpret the text surrounding a graph while incorrectly describing the plotted trend, or identify a satellite image as showing a crop type that is not visually supported.

A complete future taxonomy must therefore include **multimodal grounding hallucination**.

7.7 Evaluation Contamination and Benchmark Saturation

Static benchmarks eventually become part of training data or model-development pipelines. This can produce artificially strong performance without corresponding improvement in general reliability.

Future evaluations should therefore include:

- temporally held-out data,
- unpublished test sets,
- adversarial examples,

- dynamically changing knowledge,
- domain-shifted evidence, and
- citation-level verification.

8. Research Gaps and Future Directions

8.1 A Standardized Multidimensional Hallucination Benchmark

Current benchmarks frequently focus on one dimension. Future datasets should simultaneously annotate:

$$H = (H_F, H_{Faith}, H_C, H_T, H_R, H_U),$$

where:

- H_F = factual error,
- H_{Faith} = faithfulness error,
- H_C = citation error,
- H_T = temporal error,
- H_R = reasoning error,
- H_U = uncertainty/calibration error.

Each generated claim should ideally receive multiple labels rather than a single binary hallucination label.

8.2 Claim-Level Scientific Verification

Future research should move from document-level evaluation toward claim-level evidence graphs.

A research answer could be represented as:

$$G = (C, E, S),$$

where:

- C = generated claims,
- E = relationships between claims, and
- S = supporting sources.

Each claim should be associated with one or more evidence nodes.

This would enable automatic detection of:

- unsupported claims,
- contradictory claims,
- weak evidence,

- citation mismatch, and
 - unsupported inference chains.
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8.3 Evidence Quality Should Be Explicitly Modeled

Not every source is equally reliable.

A future hallucination detector should distinguish:

$$\text{Evidence Quality} = f(\text{authority, recency, peer review, relevance, independence}).$$

For example, a peer-reviewed systematic review may deserve greater evidentiary weight than an anonymous web page.

This is especially important because current RAG systems can retrieve irrelevant or low-quality information and still produce apparently well-grounded answers.

8.4 Citation Completeness as a First-Class Metric

A research assistant should be evaluated not only on whether its citations are correct but also whether important claims are cited.

A possible metric is:

$$CC = \frac{\text{important claims with adequate evidence}}{\text{all important claims}}.$$

This complements citation precision.

A system with high citation precision but low citation completeness may produce a polished but poorly supported literature review.

8.5 Temporal-Robust Research Agents

Research assistants should explicitly model knowledge time.

A claim should ideally carry metadata:

$$(\text{claim, source, publication date, validity interval}).$$

This would allow the system to distinguish:

- historically true,

- currently true,
- obsolete,
- uncertain, and
- future/predicted information.

FreshLLMs demonstrates the practical importance of this direction for dynamic factual QA.

8.6 Abstention and Selective Prediction

A trustworthy research assistant should not always answer.

An optimal policy may be:

$$\pi(x) = \begin{cases} \text{Answer,} & C(x) \geq \tau \\ \text{Retrieve/Verify,} & C(x) < \tau \end{cases}$$

where $C(x)$ represents estimated confidence.

This transforms hallucination mitigation from passive detection into **decision-aware generation**.

8.7 Human-in-the-Loop Scientific AI

For high-impact research claims, human verification should remain part of the workflow.

The ideal system should highlight:

- unsupported claims,
- uncertain claims,
- conflicting sources,
- low-quality evidence,
- unusual numerical results, and
- citations requiring manual checking.

Rather than replacing researchers, the LLM should function as an **evidence navigation and synthesis assistant**.

8.8 Hallucination-Aware Scientific Agents

A future research agent could implement the following architecture:

Question → Decomposition → Retrieval → Evidence Ranking → Generation → Claim Extraction → Verification
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Such an architecture would treat hallucination as a system-level reliability problem rather than merely a model-generation problem.

9. Implications for Research Applications

The taxonomy has several practical implications.

Literature Reviews

Every factual statement should ideally be traceable to a source. The system should distinguish direct findings from synthesis and interpretation.

Thesis and Dissertation Writing

LLMs should not be treated as authoritative bibliographic databases. References, numerical results, and technical claims require independent verification.

Scientific Programming

Code generated by an LLM can contain hallucinated APIs, nonexistent library functions, or incorrect algorithm implementations. Software hallucination can therefore be considered a specialized subtype of entity/relation and procedural hallucination.

Experimental Analysis

Reported metrics should be generated from actual experimental data rather than inferred or predicted by the model.

Systematic Reviews

Evidence extraction should preserve study populations, methodologies, limitations, effect sizes, and confidence intervals rather than merely summarizing conclusions.

Research Proposal Generation

LLMs can generate plausible research gaps that are not genuine gaps. A proposed "novel" research direction may already have been investigated. Novelty claims therefore require systematic literature verification.

10. Conclusion

Hallucination in Large Language Models is not a single phenomenon but a family of reliability failures occurring at different levels of factuality, evidence grounding, provenance, reasoning, and epistemic behavior. Traditional NLG research established the distinction between intrinsic and extrinsic hallucination, while recent LLM research has expanded the conceptual space toward factuality and faithfulness. For research applications, however, these categories must be further refined because scholarly reliability

depends not only on whether an output is factually correct but also on whether its claims are supported by appropriate sources, whether citations actually substantiate those claims, whether information is temporally valid, and whether uncertainty is appropriately communicated.

This paper proposed a research-oriented taxonomy comprising ten major categories: fabricated facts, entity/relation errors, temporal hallucinations, source contradictions, context omission/distortion, unsupported inference, citation fabrication, citation-support errors, quantitative/mathematical hallucinations, and reasoning-induced or confidence hallucinations. These categories are not mutually exclusive; a single response may contain several simultaneous failure modes.

Current mitigation techniques—including RAG, citation-aware generation, FActScore, self-consistency, semantic entropy, Chain-of-Verification, and research-and-revision systems—provide meaningful progress but do not eliminate hallucination. RAG can retrieve incorrect or irrelevant evidence; self-consistency can reinforce systematic errors; citation generation can produce valid references that fail to support the associated claims; and confidence estimation can remain poorly calibrated.

The central research direction should therefore shift from asking **"Does the LLM hallucinate?"** toward a more precise question:

"Which claims are unsupported, what type of hallucination produced them, what evidence is available, how reliable is that evidence, and how confident should the system be?"

For scholarly AI, the future is consequently unlikely to be a single hallucination-free language model. A more realistic objective is a **verification-centered research architecture** in which generation, retrieval, evidence attribution, claim-level verification, uncertainty estimation, temporal validation, and human oversight operate together.

Such systems would transform LLMs from fluent text generators into more accountable research instruments—systems that not only produce information, but also expose the evidentiary basis and uncertainty underlying that information.

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Research-oriented taxonomy summary

The proposed framework can be summarized as:

LLM Hallucination =	Factual Grounding → Fabricated Facts → Entity/Relation Errors → Temporal			
	Evidence Faithfulness → Source Contradiction → Context Omission/Distortion			
	Research Provenance → Citation Fabrication → Citation-Support Errors			
	Reasoning/Epistemics → Quantitative/Mathematical Errors → Reasoning/Confid			

The principal implication is that **hallucination evaluation for research applications should become claim-level, evidence-aware, source-aware, temporally aware, and uncertainty-aware rather than relying on a single hallucination score.**